Practical No.5

Aim: Key Exchange using Diffie-Hellman:
Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

Code for implementing Diffie-Hellman Algorithm :

```
if __name__ == "__main__":
    N = 23
    G = 37

    print('The Value of N is: %d' % N)
    print('The Value of G is: %d' % G)

    a = 4
    print('Secret Number for Priya is: %d' % a)
    x = pow(G, a, N)  # Priya's public value

    b = 5
    print('Secret Number for Preet: %d' % b)
    y = pow(G, b, N)  # Preet's public value

    ka = pow(y, a, N)  # Priya computes shared key
    kb = pow(x, b, N)  # Preet computes shared key

    print('Secret Key for Priya is: %d' % ka)
    print('Secret Key for Preet is: %d' % kb)
```

Output:

```
========== RESTART: E:\INS\hellman.py =
The Value of N is: 23
The Value of G is: 37
Secret Number for Priya is: 4
Secret Number for Preet: 5
Secret Key for Priya is: 2
Secret Key for Preet is: 2
```

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